Agile Lab Fat

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Clone Repo:-

```
PS E:\> git clone https://github.com/ethical0101/Travel-Site.git
Cloning into 'Travel-Site'...
remote: Enumerating objects: 19, done.
remote: Counting objects: 100% (19/19), done.
remote: Compressing objects: 100% (14/14), done.
remote: Total 19 (delta 4), reused 18 (delta 3), pack-reused 0 (from 0)
Receiving objects: 100% (19/19), done.
Resolving deltas: 100% (4/4), done.
```

Showing Commits History :-

```
PS F:\travel-site> git log --oneline
c4de445 (HEAD -> main, origin/main, origin/HEAD) MOdified Deployment.yaml
259f005 Modified jenkinsfile
6bf3315 Added jenkins and k8s
56fa623 Initial Commit
```

Project Structure:-

```
TRAVEL-SITE
  k8s
    deployment.yaml
    service.yaml
  Dockerfile
  Jenkinsfile
  index.html
```

Index.html

<!DOCTYPE html>

```html
<html>
<head>
  <title>ABC Tours & Travels</title>
  <style>
    body { font-family: Arial; margin: 0; }
    header { background: #0077b6; color: white; padding: 15px; text-align: center; }
    nav { background: #023e8a; padding: 10px; text-align: center; }
    nav a { color: white; margin: 10px; text-decoration: none; font-weight: bold; }
    section { padding: 20px; }
    footer { background: #03045e; color: white; text-align: center; padding: 10px; }
  </style>
</head>

<body>

<header>
  <h1>ABC Tours & Travels</h1>
</header>

<nav>
  <a href="#home">Home</a>
  <a href="#destinations">Destinations</a>
  <a href="#packages">Packages</a>
  <a href="#about">About</a>
  <a href="#contact">Contact</a>
</nav>
```

```
<section id="home">
    <h2>Welcome</h2>
    <p>Explore the best travel destinations!</p>
</section>

<section id="destinations">
    <h2>Destinations</h2>
    <h3>Goa</h3>
    <p>Beaches, nightlife - ₹10,000</p>

    <h3>Ooty</h3>
    <p>Hill station - ₹8,000</p>

    <h3>Hampi</h3>
    <p>Heritage site - ₹6,000</p>
</section>

<section id="packages">
    <h2>Packages</h2>
    <ul>
        <li>Weekend - ₹5000</li>
        <li>Family - ₹12000</li>
        <li>Luxury - ₹25000</li>
    </ul>
</section>
```

```
<section id="about">
    <h2>About Us</h2>
    <p>We provide best travel services.</p>
</section>

<section id="contact">
    <h2>Contact</h2>
    <p>Email: abc@gmail.com</p>
    <p>Phone: 9876543210</p>
</section>

<footer>
    <p>© 2026 ABC Tours</p>
</footer>

</body>
</html>
```

127.0.0.1:5500/index.html

ABC Tours & Travels

Home Destinations Packages About Contact

Welcome

Explore the best travel destinations!

Destinations

Goa

Beaches, nightlife - ₹10,000

Ooty

Hill station - ₹8,000

Hampi

Heritage site - ₹6,000

Packages

- Weekend - ₹5000
- Family - ₹12000
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About Us

We provide best travel services.

Contact

Email: abc@gmail.com

Phone: 9876543210

© 2026 ABC Tours

```
PS F:\travel-site> git init
Initialized empty Git repository in F:/travel-site/.git/
PS F:\travel-site> git add .
PS F:\travel-site> git commit -m "Initial Commit"
[main (root-commit) 56fa623] Initial Commit
 5 files changed, 71 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 Jenkinsfile
 create mode 100644 index.html
 create mode 100644 k8s/deployment.yaml
 create mode 100644 k8s/service.yaml
PS F:\travel-site>
```

```
PS F:\travel-site> git remote add origin https://github.com/ethical0101/Travel-Site.git
PS F:\travel-site> git branch -M main
PS F:\travel-site> git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 1007 bytes | 251.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/ethical0101/Travel-Site.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Docker Build:-

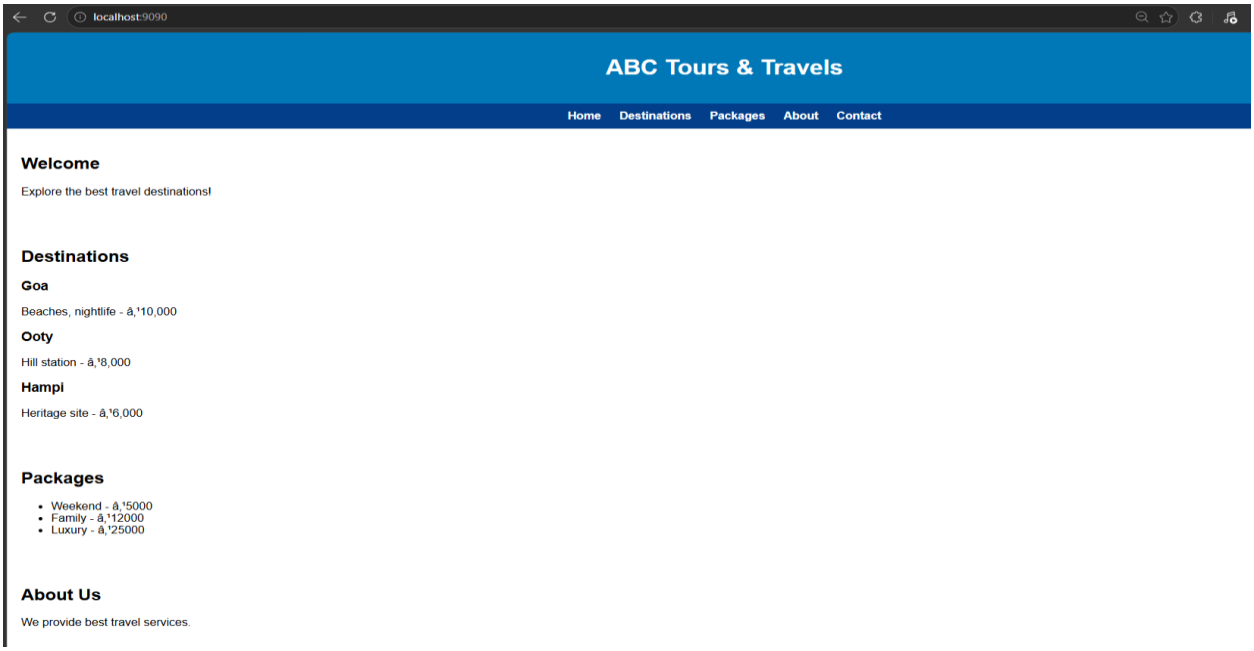
```
PS F:\travel-site> docker build -t travel-site .
[+] Building 22.0s (8/8) FINISHED                                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                0.0s
 => => transferring dockerfile: 86B                                                                                 0.0s
 => [internal] load metadata for docker.io/library/nginx:alpine                                                     2.9s
 => [auth] library/nginx:pull token for registry-1.docker.io                                                        0.0s
 => [internal] load .dockerignore                                                                                   0.0s
 => => transferring context: 2B                                                                                     0.0s
 => [internal] load build context                                                                                   0.2s
 => => transferring context: 34.28kB                                                                                0.2s
 => [1/2] FROM docker.io/library/nginx:alpine@sha256:582c496ccf79d8aa6f8203a79d32aaf7ffd8b13362c60a701a2f9ac64886c93d  18.6s
 => => resolve docker.io/library/nginx:alpine@sha256:582c496ccf79d8aa6f8203a79d32aaf7ffd8b13362c60a701a2f9ac64886c93d  0.0s
 => => sha256:1165b869c51a1a0747d78cec8fab96c30156a979e51ecf2f91aa792e557d94a4 20.25MB / 20.25MB                  18.1s
 => => sha256:c8a2fa3a88d244a3f32dcbc9c1f7649c662661a28c624198ada43aa0b7598e7f 1.40kB / 1.40kB                     0.9s
 => => sha256:34dfdd2ef1f920d0054dde2fc09ddc83ff8e71d05fadb79e2cab6e6234596f0a 1.21kB / 1.21kB                     1.3s
 => => sha256:a71873b303e8d75170b7ced2725b01b3ae15ad76f0d4eef16a49335821b6a0ef 404B / 404B                         1.3s
 => => sha256:ff9f59a6a62e9e9f29d7a84fb18865b45664d3f0d061eff7548bd61746dd101c 957B / 957B                         0.6s
 => => sha256:15e759724ff67f262e38bb7c070af9d0b84f959f9b37fa966f68bf2f881a4b62 627B / 627B                         0.6s
 => => sha256:f03becc8ac15611cfcc421c977a5ba4d65456093570788523a4ba557689aa7f7 1.87MB / 1.87MB                     3.5s
 => => extracting sha256:f03becc8ac15611cfcc421c977a5ba4d65456093570788523a4ba557689aa7f7                           0.1s
 => => extracting sha256:15e759724ff67f262e38bb7c070af9d0b84f959f9b37fa966f68bf2f881a4b62                           0.0s
 => => extracting sha256:ff9f59a6a62e9e9f29d7a84fb18865b45664d3f0d061eff7548bd61746dd101c                           0.0s
 => => extracting sha256:a71873b303e8d75170b7ced2725b01b3ae15ad76f0d4eef16a49335821b6a0ef                           0.0s
 => => extracting sha256:34dfdd2ef1f920d0054dde2fc09ddc83ff8e71d05fadb79e2cab6e6234596f0a                           0.0s
 => => extracting sha256:c8a2fa3a88d244a3f32dcbc9c1f7649c662661a28c624198ada43aa0b7598e7f                           0.0s
 => => extracting sha256:1165b869c51a1a0747d78cec8fab96c30156a979e51ecf2f91aa792e557d94a4                           0.4s
 => [2/2] COPY . /usr/share/nginx/html                                                                              0.1s
 => exporting to image                                                                                              0.2s
 => => exporting layers                                                                                             0.1s
 => => exporting manifest sha256:e196f959ecc1fc9a8a846b7903ee990b3288b7f66bb542fe51bc93a2ed4363f9                   0.0s
```

```
 => [2/2] COPY . /usr/share/nginx/html                                                                              0.1s
 => exporting to image                                                                                              0.2s
 => => exporting layers                                                                                             0.1s
 => => exporting manifest sha256:e196f959ecc1fc9a8a846b7903ee990b3288b7f66bb542fe51bc93a2ed4363f9                   0.0s
 => => exporting config sha256:2b86c5e7d0f108e08f5b8fc71ded75fee471b59d473b7044e4dd6166fc93d47e                     0.0s
 => => exporting attestation manifest sha256:7ed0ebb05e19b421efb29d3d7a9ea5dba4a78dc2a0a3f8b9d942b52924cd4af9       0.0s
 => => exporting manifest list sha256:525aa69fa6c936a3291efd948df64cde7e71de75dc0e3f59c821f4d0a43f64c1              0.0s
 => => naming to docker.io/library/travel-site:latest                                                               0.0s
 => => unpacking to docker.io/library/travel-site:latest                                                            0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/plbojx9v25njd6190oqmxkl5e
```

Running the docker image:-

```
PS F:\travel-site> docker run -d -p 9090:80 travel-site
fe756fa71abdf8c33515e1c2123732a5fef9fd4fe9a6ea255145ad62acec2cb7
```

```
PS F:\travel-site> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS         PORTS                                      NAMES
fe756fa71abd   travel-site    "/docker-entrypoint.…"   4 minutes ago   Up 4 minutes   0.0.0.0:9090->80/tcp, [::]:9090->80/tcp    wizardly_kapitsa
```



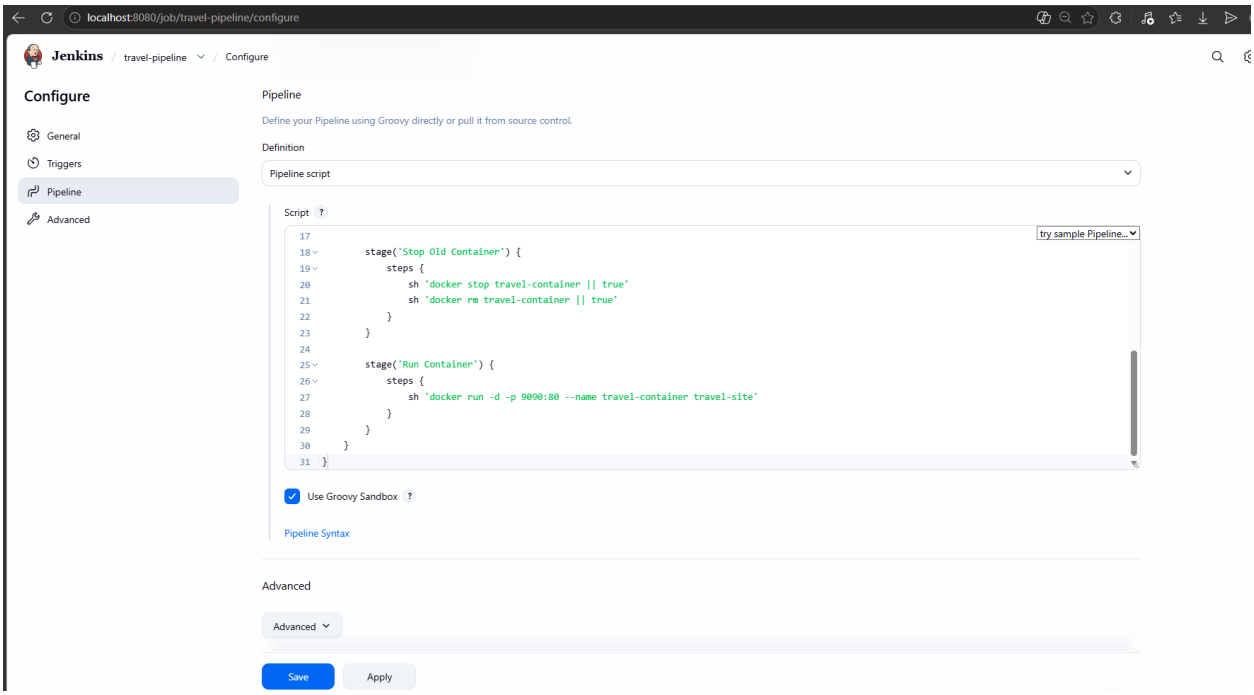
Configuring Jenkins:-

pipeline code:-

```
pipeline {
   agent any

   stages {

      stage('Clone Code') {
         steps {
            git branch: 'main', url: 'https://github.com/ethical0101/Travel-Site.git'
         }
      }

      stage('Build Docker Image') {
```

```
        steps {
            bat 'docker build -t travel-site .'
        }
    }

    stage('Stop Old Container') {
        steps {
            bat 'docker stop travel-container || exit 0'
            bat 'docker rm travel-container || exit 0'
        }
    }

    stage('Run Container') {
        steps {
            bat 'docker run -d -p 9090:80 --name travel-container travel-site'
        }
    }
}
```

}



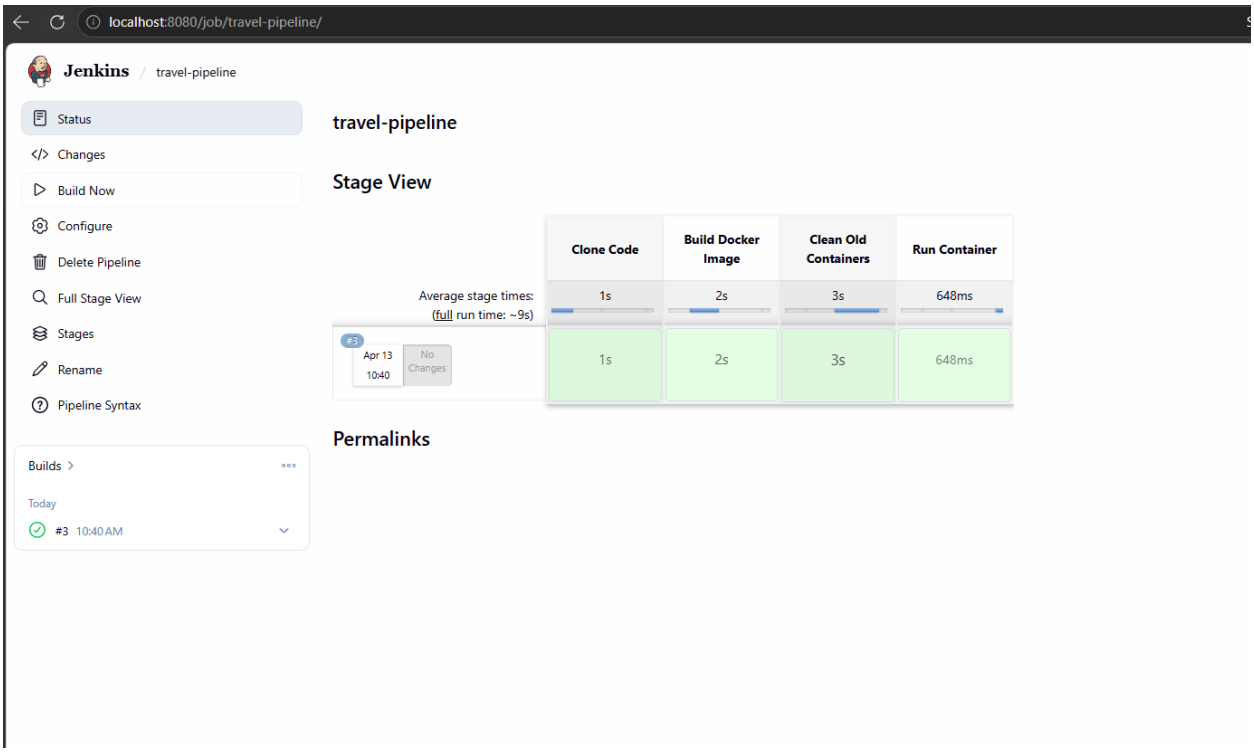
deployment.yaml:-

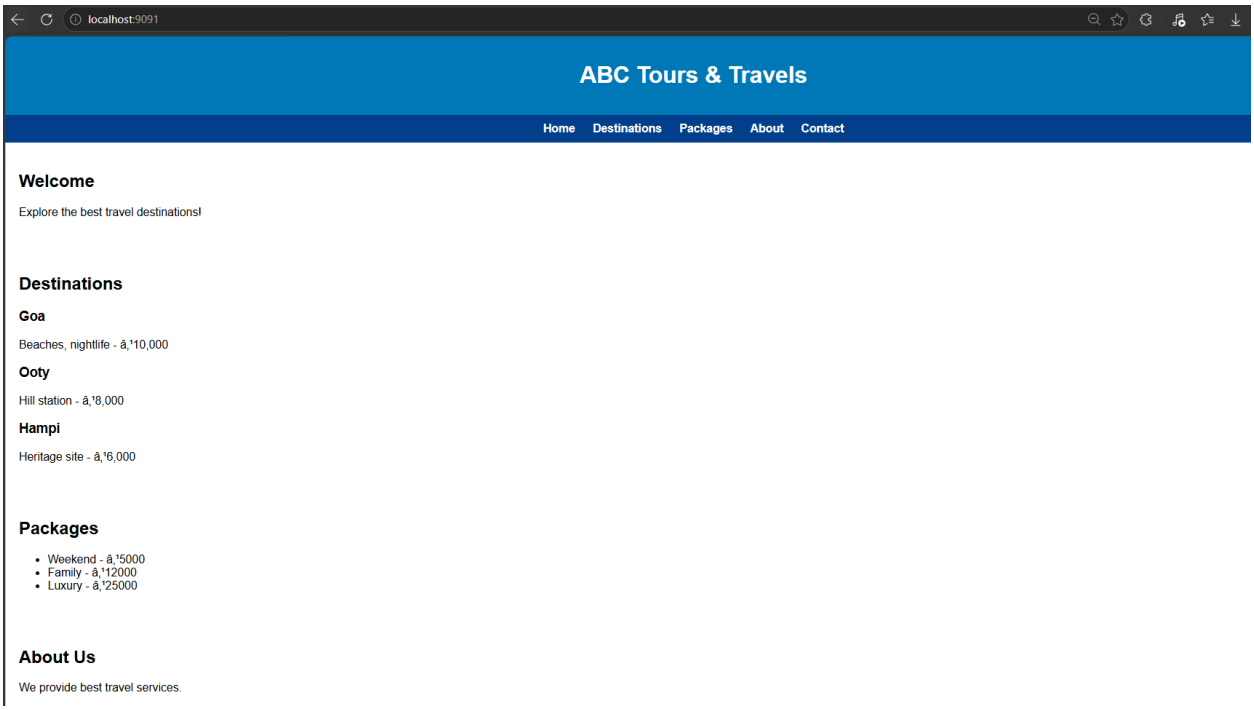
```
apiVersion: apps/v1

kind: Deployment

metadata:

  name: travel-app

spec:

  replicas: 2

  selector:

    matchLabels:

      app: travel

  template:

    metadata:

      labels:

        app: travel
```

```
  spec:
    containers:
    - name: travel-container
      image: travel-site
      ports:
      - containerPort: 80
```

service.yaml:-

```
apiVersion: v1
kind: Service
metadata:
  name: travel-service
spec:
  type: NodePort
  selector:
    app: travel
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30007
```



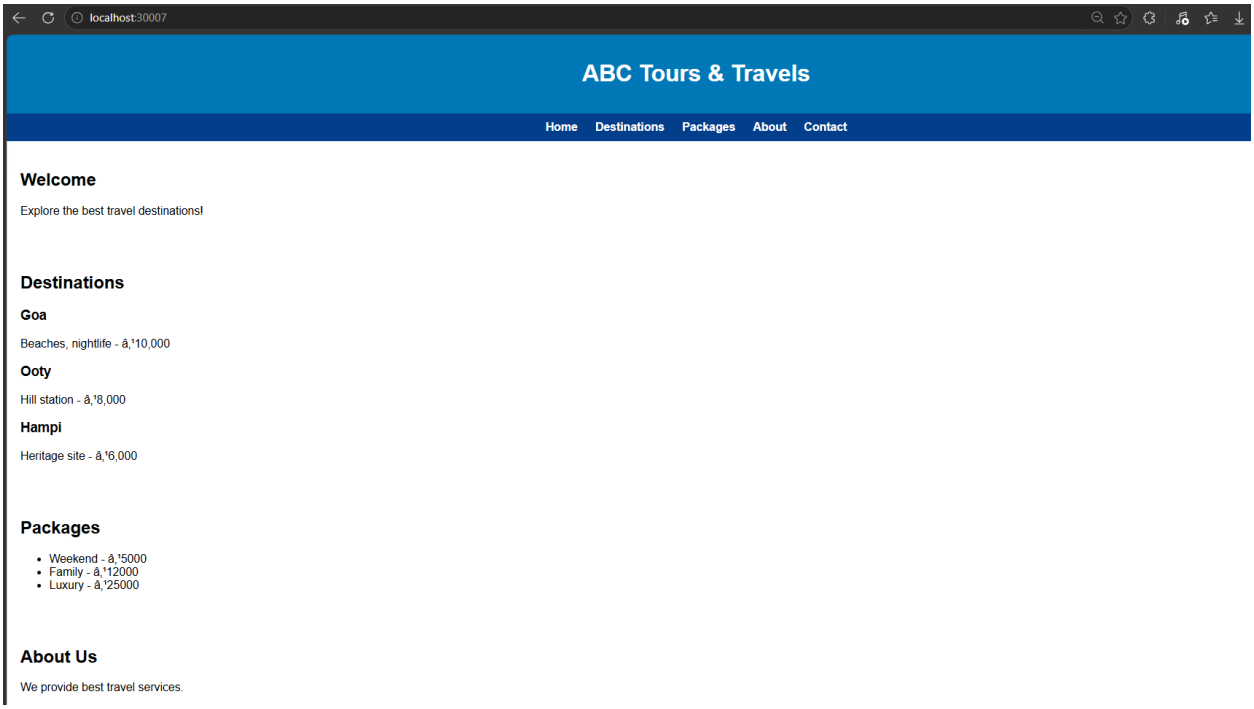


Docker config:-

```
PS F:\travel-site> docker login
i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
PS F:\travel-site> docker tag travel-site druthendra/travel-site
PS F:\travel-site> docker push druthendra/travel-site
Using default tag: latest
The push refers to repository [docker.io/druthendra/travel-site]
3fab724022e8: Pushed
a71873b303e8: Pushed
c8a2fa3a88d2: Pushed
589002ba0eae: Pushed
34dfdd2ef1f9: Pushed
ff9f59a6a62e: Pushed
1165b869c51a: Pushed
f03becc8ac15: Pushed
15e759724ff6: Pushed
a4392aaa3f7e: Pushed
latest: digest: sha256:59afa6d0131d22bcae0095e4b96026f804415490f9ce74f750ee4c4077a764b8 size: 856
PS F:\travel-site>
```

Applying Kubernetes:-

```
PS F:\travel-site> kubectl apply -f k8s/deployment.yaml
deployment.apps/travel-app created
PS F:\travel-site> kubectl apply -f k8s/service.yaml
service/travel-service created
PS F:\travel-site>
```

```
PS F:\travel-site> kubectl get pods
NAME                          READY   STATUS    RESTARTS        AGE
html-demo-5d78575d7c-2kk4w    1/1     Running   4 (5m10s ago)   18d
html-demo-5d78575d7c-xn85f    1/1     Running   4 (5m10s ago)   18d
travel-app-5b7c6985fc-pzlrq   1/1     Running   0               28s
travel-app-5b7c6985fc-xqmqv   1/1     Running   0               28s
PS F:\travel-site> kubectl get svc
NAME                TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
html-demo-service   NodePort    10.97.141.209   <none>        80:30010/TCP   18d
kubernetes          ClusterIP   10.96.0.1       <none>        443/TCP        18d
travel-service      NodePort    10.99.118.64    <none>        80:30007/TCP   32s
PS F:\travel-site>
```

Accessing the web page at port 30007